

App Development Project Report

App Dev Project Report

Student Details

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About Me: I am a Placement cell Intern at IIT Madras BS Degree program with a deep interest in robotics and model architecture. I enjoy building and breaking stuff.

Project Title

Placement Portal System

Project Overview

The Placement Portal Management System is a web-based application developed to manage campus placement activities. It allows students to register, upload resumes, view placement drives, and apply for jobs. Companies can register, create placement drives, and manage student applications. The admin manages students, companies, departments, approvals, drives, and application records.

Approach

The app was built using Flask as the backend framework with first principles thinking making it basic ui and clean and fresh looking and totally operation oriented.

AI/LLM Declaration

I used chatGPT to understand topics and to generate the css files for styling and some backend logic also which i was not getting. The extent of AI/LLM usage is around 10–12%, limited to code suggestions and documentation formatting

Technology Used

Area	Technology
Backend	Python Flask
Frontend	HTML, CSS
Database	SQLite
File Upload	Resume upload stored in static/resumes
Templates	Flask Jinja Templates

Main Users

Admin

Logs into admin dashboard
Approves or rejects companies
Approves or rejects placement drives
Manages students, companies, and departments
Views all applications

Student

Registers and logs in
Updates profile and resume
Views available placement drives
Applies for drives
Tracks application status

Company

Registers and logs in
Updates company profile
Creates placement drives
Views student applications
Updates application status and remarks

Main Modules

Module	Description
Student Module	Handles student registration, login, profile update, resume upload, and drive applications
Company Module	Handles company registration, login, profile update, drive creation, and application review
Admin Module	Handles approvals, account status management, department management, and dashboard monitoring
Drive Module	Stores placement drive details created by companies
Application Module	Stores student applications for placement drives
Department Module	Stores department names used during student registration

Database Tables

Table	Purpose
admins	Stores admin login details
students	Stores student profile and academic details
companies	Stores company profile and approval details
departments	Stores department names
drives	Stores placement drive/job details
applications	Stores student applications for drives

ER Diagram Explanation:

- One Company can create many Drives.
- One Drive belongs to one Company.
- One Student can submit many Applications.
- One Drive can receive many Applications.
- One Application connects one Student with one Drive.
- One Department can have many Students.
- Admin is separate because admin manages the system but does not directly belong to the main placement application flow.

ER Diagram

ADMINS		
INTEGER	id	PK
TEXT	username	
TEXT	password	
TEXT	full_name	

DEPARTMENTS		
INTEGER	id	PK
TEXT	name	
TEXT	created_at	

COMPANIES		
INTEGER	id	PK
TEXT	name	
TEXT	username	
TEXT	password	
TEXT	email	
TEXT	phone	
TEXT	website	
TEXT	details	
TEXT	approval_status	
TEXT	account_status	
TEXT	created_at	

STUDENTS		
INTEGER	id	PK
TEXT	roll_number	
TEXT	name	
TEXT	username	
TEXT	password	
TEXT	email	
TEXT	phone	
TEXT	department	
TEXT	study_year	
TEXT	gender	
TEXT	cgpa	
TEXT	resume_path	
TEXT	account_status	
TEXT	created_at	

DRIVES		
INTEGER	id	PK
INTEGER	company_id	FK
TEXT	title	
TEXT	job_role	
TEXT	description	
TEXT	location	
TEXT	salary	
TEXT	drive_date	
TEXT	last_date_to_apply	
TEXT	approval_status	
TEXT	lifecycle_status	
TEXT	created_at	

APPLICATIONS		
INTEGER	id	PK
INTEGER	drive_id	FK
INTEGER	student_id	FK
TEXT	status	
TEXT	remark	
TEXT	applied_at	

has

creates

submits

receives

Architecture

This is a monolithic web application architecture.

All main parts of the application are inside one Flask project:

Layer	Files / Folder	Purpose
Presentation Layer	templates/, static/	Displays pages using HTML and CSS
Application Layer	portal_app.py	Handles routes, login, validation, dashboard logic, and form actions
Data Layer	placement_portal.db	Stores students, companies, drives, applications, departments, and admins
Entry Point	app.py	Starts the Flask application

Video Presentation

Link -

<https://drive.google.com/file/d/1PpPpyDE4hWDIV884ioJ0USFgOUCFHRxa/view?usp=sharing>